Dr. KAUSHIK NAYAK (F-162)

Associate Professor, Dept. of Electrical Engineering;
Adjunct Faculty, Dept. of Engineering Sciences;
Indian Institute of Technology Hyderabad, India.
Senior Member (IEEE)
URL: https://www.iith.ac.in/~knayak/

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1. **D.O.B:** 29-05-1980; **Gender:** Male; **Category:** General
2. **Email ID and contact number(s):**

E-mail: knayak@ee.iith.ac.in;

Contact: +91 99854 39940, 94934 36940

3. **Research Interests (Core):** Electron Devices Physics (Theory, Modeling & Simulations), Nanoscale Transistor/Logic Devices Physics, Mesoscopic Electronics.

- **Other Intellectual Interests:** History and Philosophy of Science, Earth System Science, Planetary Sciences & Cosmology.

4. **Academic Qualifications:**

Sl. No.	Degree	Year	Specialization/ Department	University
1	Ph.D.	2014	Nano-electronic Devices, Dept. of Electrical Engineering	IIT Bombay
2	M. Tech.	2007	Microelectronics, Dept. of Electrical Engineering	IIT Bombay
3	B.E.	2001	Electronics and Telecommunication Engineering	Utkal University

5. Work experience (in reverse chronological order):

Sl. No.	Position Held	Institute/Organization	From	To
1	Associate Professor (current)	Dept. of Electrical Engineering (IIT Hyderabad)	13th October 2021	Till Date
2	Assistant Professor	Dept. of Electrical Engineering (IIT Hyderabad)	3rd August 2015	12th October 2021
3	Adjunct Faculty Member	Dept. of Climate Change (IIT Hyderabad)	August 2019	December 2024
4	Adjunct Faculty Member	Dept. of Engineering Science (IIT Hyderabad)	September 2022	Till Date
5	Senior Device Engineer	Global Foundries, Singapore	19 January 2015	9 March 2015
6	Ph.D. Research Intern	IBM Semiconductor Research and Development Centre	March 2012	March 2014
7	Global COE Program Visiting Scholar	Frontier Research Centre, Tokyo Institute of Technology	8 February 2010	27 February 2010
8	Assistant Professor	Dept. of ECE, ITER, SOA University, Bhubaneswar	21 July 2007	10 July 2008
9	Lecturer	Dept. of ECE, ITER, SOA University, Bhubaneswar	01 June 2002	20 July 2007

6. Academic Courses Offered in Total (Teaching endeavors at IIT Hyderabad in the last decade): 14 Courses in Electrical Engineering Department and 3 Courses in Climate Change Department as Adjunct faculty.

EE 5110/5181 *(Semiconductor Devices and Modeling)- 3 credits (PG);*
EE 4110 *(Physics of Electrical Engineering Materials) – 3 credits (UG)****;***
EE 6180 *(Semiconductor Heterojunction Devices Physics) – 3 credits (PG);*
EE 6170 *(Mesoscopic Device Electronics) – 3 credits (PG);*
EE 4170 *(Mesoscopic Device Electronics) – 3 credits (UG);*
EE 6440 *(Extreme Environment Physical Electronics) – 3 Credits (PG);*
EE 6160 *(Mesoscopic Carrier Transport) – 2 credits;*
EE 5107*(Semiconductor Physical Electronics) – 2 credits;*
EE 5117 *(Microelectronic Device Physics) – 1 credit;*
EE 5111/5189 *(Microelectronics and Device Simulation Laboratory) – 2 credits;*
EE 2330 *(Antenna Design) – 1 credit;*
EE 3010 (*Electromagnetic Waves and Transmission Lines) – 2 credits;*
EE 3302 *(Electromagnetic Wave Propagation) – 3 credits;*
EE 6120 *(Nanoelectronics Principles and Devices) – 3 credits;*

CC 5110 *(Earth's Climate and Atmospheric Sciences) – 3 credits;*
CC 5130 *(Atmospheric Electricity) – 3 credits;*
CC 5010 *(Earth's Evolution and Paleoclimate) – 1 credit.*

7. New Academic Courses Proposed & Offered (After becoming Associate Professor, EE): 4

EE 4110 *(Physics of Electrical Engineering Materials) – 3 credits (UG)****;***
EE 6180 *(Semiconductor Heterojunction Devices Physics) – 3 credits (PG);*
EE 6440 *(Extreme Environment Physical Electronics) – 3 Credits (PG);*
EE 4170 *(Mesoscopic Device Electronics) – 3 credits (UG);*

8. Research Publications Profile: (Total Int. Journals: 18; Total Conference Proc. :10)

(Separate Sheet is attached with details)

Google Scholar: h-index: 14; i10-index: 16; Citations: 692	**Scopus ID: 21743391200; h-index: 13; Citations: 581**

9. **Supervised & Graduated EE Ph.D. Scholars (*Total 4*) (Reverse Chronological Order)**

- **(During current position of Associate Professor): 2**

[i] Yerragudi Pullaiah (EE17RESCH11010): **(**Supervisor: Kaushik Nayak) – **Thesis:** ***Analysis and Performance Exploration of Diamond Based Unipolar Devices Under High Power & Extreme Environment Conditions*** (*Ph.D. Defense Viva-voce: 19 December 2024*).

[ii] Kumar Prashant (EE16RESCH11010): **(**Supervisor: Kaushik Nayak) – **Thesis:** ***Engineering Electrical Contacts to 2-D Transition-Metal Dichalcogenide (TMDC) FETs*** (*Ph.D. Defense Viva-voce: 12th April 2023*).

- **(During position of Assistant Professor): 2**

[iii] Akhil Sudarsanan (EE15RESCH11004): **(**Supervisor: Kaushik Nayak)– **Thesis:** ***Variability Aware CMOS Device Design and Simulations of sub-10 Confined Geometry Nano-MOSFETs*** (*Ph.D. Defense Viva-voce: 20th July 2021*).

[iv] Sankatali Venkateswarlu (EE15RESCH11007): (Supervisor: Kaushik Nayak) – **Thesis:** ***Electro- thermal Aware Device Design and Simulations of Confined Geometry Nano-MOSFETs for Sub-10nm CMOS Applications*** (*Ph.D. Defense Viva-voce: 4th November 2020*).

10. **Ongoing EE Ph.D. Scholar supervision**: **1** full-time Ph.D. Scholar (Admitted in July 2025): *Mani Ram (EE25RESCH11006) – DST Inspire Fellow* (*Tentative Topic: Logic Device Modeling & Simulation of Nanosheet FET based C-FETs for Angstrom era 3D Integration).*

11. **M. Tech. (Microelectronics) Thesis Supervised:*(6 Completed)* 4 students** as Individual Guide and **2 students** as Co-Guide; **1 student Thesis supervision ongoing** (Since March 2025 -).

12. **Research Laboratory Established in EE, IIT Hyderabad (Since 2016):**

Electron Devices Research (EDR) Laboratory (Device Physics & Engineering: Theory, Modeling, Design and Simulations): *500 SQ FT laboratory housing high-end computer servers and systems for Devices Physics research.*

Location: Laboratory Room# EEL-34, Department of Electrical Engineering, IIT Hyderabad.

13. Externally Funded and/or Collaborative Research Projects as P.I.:

Sl. No.	Name of Agency	Title of Research project	Co-PI(s) from IITH	Total Amount	Period	Completed/on-going
1	Department of Science And Technology (DST), Govt. of India.	Development of Ternary Organic solar cells on low cost flexible substrates.	Prof. Shiv Govind Singh	INR 49,35,000 /-	2017-2020	Completed
2	Synopsys Inc.	Diamond m-i-p+ Schottky Diode Characteristics Modeling including the effects of surface states CNL and MIGs	Dr. Oves Badami	NA	2021 - 2022	Completed
3	Synopsys Inc.	Analysis and Performance Exploration of Diamond Based Unipolar Devices Under High Power & Extreme Environment Conditions	NA	NA	2020-2024	Completed
4	Synopsys Inc.	Lateral Diamond Schottky Barrier Diode Modeling & Device	NA	NA	2025 -	Ongoing

		Analysis for High Breakdown Voltage				
5	Synopsys Inc.	Electro-thermal Aware Device Design and Analysis of Nano-sheet based CFETs for Angstrom length Logic Technology	NA	NA	2025-	Ongoing

14. **Administrative Support Activities at IIT Hyderabad:**

[i] **IIT Hyderabad Library Committee Member** *(Knowledge Resource Centre):*

- Representing Electrical Engineering Department: *5th March 2025 – till date.*
- Earlier Membership (EE Dept.): *25th January 2018 – 9th February 2023.*

[ii] **EE Departmental Faculty Representative (in Sunshine):**
Institute Psychological Counselling Cell **(Sunshine)-(**1st October 2020 - 14 September 2023).

[iii] **Department Representative in** Institute **Student Grievance Cell (SGC)** - Academic and Non-Academic matters:

- **Climate Change Dept.:** *March 2021 – Dec 2024.*
- **Greenko school of Sustainability:** *Aug 2024 – Dec 2024.*

[iv] **Academic Faculty Advisor (UG):** Graduated B. Tech. (Electrical Engineering) Batch: *July 2016 – July 2020.*

[v] **Academic Faculty Advisor (PG):** M. Tech. Microelectronics and VLSI (July 2021 Batch): *July 2021 – June 2023.*

[vi] Electrical Engineering Department UG Program Committee (DUGC) member: *(July 2016 – July 2020).*

[vii] Convener of Electrical Engineering Department Ph.D. Admission Committee:

(July 2018 Admission; Jan 2019 Admission).

- *EE Microelectronics and VLSI Specialization Ph.D. & M. Tech. Admission Coordinator: Multiple times since July 2016.*

[viii] Member Faculty Search Committee (FSC), EE Department – *Rounds 16 and 17: 2022 – 2023.*

[ix] Successfully coordinated and led single-handedly a team of 29 IIT Hyderabad students *(Ph.D., M. Tech., M.Sc., and B. Tech. students) visit to Japan (Tokyo, Osaka and Kobe) for a week (from 20 -26 January 2016) under JENESYS 2015 program conducted by Japan International Cooperation Center.*

15. **Professional Society Associations:** ***Senior Member IEEE (Since February 2020)***; IEEE Electron Devices Society, International Roadmap for Devices and Systems.

16. **Peer Publications Review Services: *(International Journals, Letters, Books & Conference Proceedings):***
    - *Journals/Letters: IEEE EDS Journals: IEEE Transactions on Electron Devices, IEEE Electron Devices Letters, Journal of the Electron Devices Society, IEEE Transactions on Device and Materials Reliability, IEEE Journal of Exploratory Solid-State Computational Devices and Circuits; AIP Journal of Applied Physics; Elsevier Applied Mathematical Modelling; Electronics Letters (IET & Wiley); Springer Nature: Asia Materials, Springer Journal of Computational Electronics; Japanese Journal of Applied Physics (JJAP), etc.*
    - *Books: Springer Publication Books review.*
    - *Conference Papers: IEEE International Conference on Emerging Electronics (ICEE) (2022, 2025)- Logic Devices Track; IEEE EDTM 2024- Logic Devices Track; International Workshop on Physics of Semiconductor Devices (IWPSD 2019, 2021); International Symposium on VLSI Design and Test (2020); International Conference on Solid State Devices and Materials (SSDM) (2024, 2025).*

17. Other Outreach Activities & External Professional Services:

- *External Thesis Review & Examination: IIT Palakkad (PhD Thesis), IIT Gandhinagar (PhD Thesis), NIT Kurukshetra (PhD Thesis), IIT Madras (MS Thesis), University of Hyderabad (M. Tech. Thesis).*
- *Prime Minister's Research Fellowship (PMRF) Selection Expert Committee Member (2021): Selecting candidates for the Electrical Engineering and Electronics Engineering discipline (Coordinated by IIT Madras).*
- *Technical Committee member – Logic Devices Track - IEEE International Conference on Emerging Electronics (ICEE) (2020, 2022, 2025); IEEE EDTM 2024.*
- *Technical Committee member - Logic Vertical (2025), Development of Semiconductor Ecosystem Talent (DevSET) - Ministry of Education, GoI (Mentor: IIT Bombay)*

18. Invited Talks/Expert Lectures (Reverse Chronological Order):

[i] Lecture talk on *"Historical Evolution of CMOS Technology,"*- Scheme for Promotion of Academic and Research Collaboration (SPARC) Workshop, Dept. EE, IIT Hyderabad (3rd June 2025 & 6th Dec 2024).

[ii] Invited talk on *"Extreme Environment Resistance Investigation of H-terminated Diamond MOSFET,"* – Faculty Development Program (FDP), Dept. of ECE, NIT Raipur (23-04-2025).

[iii] Invited talk on *"Dawn of Semiconductor and Device Electronics (A Historical Survey),"*- Dept. of Electronics Engineering, G H Raisoni College of Engineering Nagpur (25-09-2024).

[iv] Invited Talk on *"Exploring Semiconductor Surface States Physics in Diamond Schottky Diode,"*- AICTE - STTP Workshop, Dept. of ECE, NIT Calicut (27-07-2023).

[v] Invited talk on *"Diamond Electron Devices for Extreme Environment Electronics,"*- 6th IEEE ICEE, Hilton, Bengaluru, (13-12-2022).

[vi] Invited Lecture Talk on *"Extreme Environment Electron Devices Engineering,"*- Dept. of ECE, GIET University, Gunupur, (11-07-2022).

[vii] Invited talk in Short-Term Program on *"Emerging Nanoscale Devices, Circuits and Their Applications,"*- Department of Electronics and Communication Engineering, Delhi Technological University, scheduled May 10-14, 2021.

[viii] Invited talk on *"Device Self-heating Effects in sub-deca-nanometer Logic Transistors,"*- AICTE- sponsored short term training program (STTP), organized by VNRVJIET, Hyderabad, (05-03-2021).

[ix] Invited talk on *"Wide Bandgap Semiconductors for Energy Efficient Electronic Devices,"* 5th International Conference on Reuse and Recycling of Materials and their products (ICRM – 2020), Jointly Organized by Mahatma Gandhi University, Kottayam (India) & Wroclaw University of Technology (Poland), 11 - 13 Dec 2020.

[x] Invited talk on *"O-terminated Diamond (D:O) MOS Devices for High Temperature and High Voltage Applications,"*- 5th International conference on Emerging Electronics (IEEE-ICEE 2020), IIT Delhi, 26th - 28th Nov. 2020.

[xi] Invited talk on *"O-terminated p-Diamond (D:O) MOSFET Device Electrostatics and High Temperature Performance," -* International Symposium on Semiconductor Materials and Devices (ISSMD-2020), Dr. B. R. Ambedkar National Institute of Technology, Jalandhar, 31st Oct - 2nd Nov 2020.

[xii] Invited talk on "*Nanometer-scale MOSFETs in CMOS Nanoelectronics,"-* Webinar organized by IEEE Circuits and Systems Society Chapter of Vaagdevi College of Engineering, Warangal, 8 June 2020.

[xiii] Invited talk on *"Nanometer-scale CMOS Logic Transistors (Historical Trends and Future scope),"* Workshop on Recent trends in Materials and its applications, CBIT, Hyderabad, 20 Nov 2019.

[xiv] Invited talk on *"History and Evolution of Solid-State Electronic Devices,"* Guest Lecture VNR Vigyana Jyothi Institute of Engineering & Technology, Hyderabad, 18 Aug 2017.

I hereby declare that the information provided above is true to the best of my knowledge and belief.

(Dr. Kaushik Nayak)

List of References:

- **Prof. V. Ramgopal Rao**,
Fellow of IEEE, TWAS, INAE, INSA, IASc, NASI
Vice-Chancellor, Birla Institute of Technology & Science
(Pilani, Hyderabad, Goa, Dubai & Mumbai)
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(More References available upon request)

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URL: https://www.iith.ac.in/~knayak/

Complete List of Research Publications: (18 International Journals and 10 International Conference papers)

I. *List of Publications after joining as Associate Professor in EE Dept., IIT Hyderabad (Based on the work carried out with PhD Students after 13 Oct 2021): (Reverse Chronological Order)*

[1] Yerragudi Pullaiah, Mohit Bajaj, and **Kaushik Nayak**, "Investigation of X-ray Irradiance Hardness and High Temperature Operation of H-Terminated Diamond MOSFET", *in **IEEE Transactions on Electron Devices,** vol. 72, no.2, pp. 557-563, February 2025.* ***DOI:*** 10.1109/TED.2024.3514813.

[2] Kumar Prashant and **Kaushik Nayak**, "Contact Analysis of Elemental Transition Metal Electrodes for Complementary 2D-FET applications using MoS2 and WSe2", *in **IEEE Electron Device Letters,** vol.43, Issue 7, pp. 1137 - 1140, July 2022.* ***DOI:*** *10.1109/LED.2022.3180083.*

[3] Yerragudi Pullaiah, Mohit Bajaj, Oves Badami, and **Kaushik Nayak**, "TCAD Analysis of O-terminated Diamond m-i-p+ Diode Characteristics Dependencies on Surface States CNL and Metal-Induced Gap States", *in **IEEE Transactions on Electron Devices**, vol. 69, no. 1, pp. 271 - 277, January 2022.* ***DOI:*** 10.1109/TED.2021.3129726.

[4] Kumar Prashant, Dinesh Gupta, Yerragudi Pullaiah, Oves Badami, and **Kaushik Nayak**, "Electrode orientation dependent transition metal—(MoS2 ; WS2) contact analysis for 2D material-based FET applications", *in **IEEE Electron Device Letters**, vol.42, Issue 12, pp. 1878-1881, December 2021.* ***DOI:*** 10.1109/LED.2021.3121810.

II. List of Publications before joining as Associate Professor in EE Dept., IIT Hyderabad (Based on the work carried out with PhD Students as Assistant Professor): (Reverse Chronological Order)

[5] Sankatali Venkateswarlu, Oves Badami, and **Kaushik Nayak,** "Electro-Thermal Performance Boosting in Stacked Si Gate-All-Around Nanosheet FET with Engineered Source/Drain Contacts", *in* ***IEEE Transactions on Electron Devices****, vol. 68, no. 9, pp. 4723 - 4728, September 2021.* **DOI:**10.1109/TED.2021.3095038.

[6] Akhil Sudarsanan, Oves Badami and **Kaushik Nayak**, "Superior Interface Trap Variability Immunity of Horizontally Stacked Si Nanosheet FET in Sub-3-nm Technology Node," ***International Semiconductor Conference (CAS), Romania***, 2021, pp. 161-164, **DOI**: 10.1109/CAS52836.2021.9604183.

[7] Akhil Sudarsanan and **Kaushik Nayak**, "Immunity to random fluctuations induced by interface trap variability in Si gate-all-around n-nanowire field-effect transistor devices", *in* ***Journal of Computational Electronics, Springer****, April 2021.* ***DOI:*** https://doi.org/10.1007/s10825-021-01692-w.

[7] Sankatali Venkateswarlu, and **Kaushik Nayak**, "Device SHEs in the Presence of Non- equilibrium Channel Heat Transport in SOI and SOD FinFETs with Technology Scaling", ***5th IEEE International Conference on Emerging Electronics (ICEE)***, New Delhi, India, 2020, pp. 1-4, **DOI:** 10.1109/ICEE50728.2020.9777080.

[8] Sankatali Venkateswarlu, and **Kaushik Nayak**, "Hetero-Interfacial Thermal Resistance Effects on Device Performance of Stacked Gate-All-Around nano-sheet FET"***, in IEEE Transactions on Electron Devices,*** vol. 67, no. 10, pp. 4493–4499, October 2020. **DOI:** 10.1109/TED.2020.3017567.

[9] Kumar Prashant, Yerragudi Pullaiah, Dinesh Gupta, and **Kaushik Nayak**, "Atomistic Modeling to Engineer Ohmic Contacts between Monolayer MoS2 and Transition Metals", ***in proceedings of IEEE International Interconnect Technology conference (IITC) 2020*, San Jose, California, USA, October 5-8, 2020.**
DOI: 10.1109/IITC47697.2020.9515662.

[10] Sankatali Venkateswarlu, and **Kaushik Nayak**, "Ambient Temperature Induced Device Self-heating Effects on Multi-Fin Si CMOS Logic Circuit Performance in N-14 to N-7 scaled Technologies", ***in IEEE Transactions on Electron Devices*,** vol. 67, no. 4, pp. 1530- 1536, April 2020. **DOI:** 10.1109/TED.2020.2975416.

[11] Akhil Sudarsanan, Sankatali Venkateswarlu, and **Kaushik Nayak**, "Superior Work Function Variability Performance of Horizontally Stacked Nanosheet FETs for Sub-7-nm Technology and Beyond", ***4th IEEE Electron Devices Technology and Manufacturing conference (EDTM) 2020*,** Penang, Malaysia, 6-21 April 2020. **DOI:** 10.1109/EDTM47692.2020.9117974.

[12] Yerragudi Pullaiah, Naresh Kumar Emani, and **Kaushik Nayak,** "Device Electrostatics and High Temperature Operation of Oxygen Terminated Boron Doped Diamond MOS Capacitor and MOSFET", ***4th IEEE Electron Devices Technology and Manufacturing conference (EDTM) 2020*** **,** Penang, Malaysia, 6-21 April 2020. **DOI:** 10.1109/EDTM47692.2020.9117884.

[13] Jinal Tapar, Saurabh Kisen, Kumar Prashant, **Kaushik Nayak,** and Naresh Emani, "Enhancement of the optical gain in GaAs nanocylinders for nanophotonic applications," AIP ***Journal of Applied Physics*,** vol. 127, no. 15, April 2020. **DOI:** https://doi.org/10.1063/1.5132613.

[14] Sankatali Venkateswarlu, Akhil Sudarsanan, and **Kaushik Nayak**, "Impact of Phonon Boundary Scattering on Self-heating Effects in Stacked Si Nano-sheet FET in sub-7nm Logic Technologies", ***20th International Workshop on Physics of Semiconductor Devices: IWPSD*,** *Kolkata, India,* Dec 17-20, 2019.

[15] Akhil Sudarsanan, Sankatali Venkateswarlu, and **Kaushik Nayak**, "Impact of Fin Line Edge Roughness and Metal Gate Granularity on Variability of 10 nm node SOI n-FinFET", ***in IEEE Transactions on Electron Devices*,** vol. 66, no. 11, pp. 4646–4652, November 2019**. DOI:** 10.1109/TED.2019.2941896.

[16] Sankatali Venkateswarlu, Akhil Sudarsanan, and **Kaushik Nayak**, "Improved Electro- Thermal Performance in FinFETs using SOD Technology for 7nm node High Performance logic Devices", ***Extended Abstracts of the 2019 International Conference on Solid State Devices and Materials, Nagoya, 2019, pp669-670.*** **DOI:** https://doi.org/10.7567/SSDM.2019.PS-1-23.

[17] Jinal Tapar, Saurabh Kisen, **Kaushik Nayak**, and Naresh Emani, "Optimizing the Gain in Semiconductor Nanostructures for All-dielectric Active Metamaterial Applications", **10th** ***International Conference on Materials for Advanced Technologies (ICMAT)***, Marina Bay Sands, Singapore, 23-28 June 2019.

[18] Sankatali Venkateswarlu, Akhil Sudarsanan, Shiv Govind Singh, and **Kaushik Nayak**, "Ambient Temperature Induced Device Self-heating Effects on Multi-Fin Si n-FinFET Performance," ***in IEEE Transactions on Electron Devices***, vol. 65, no. 7, pp. 2721 – 2728, July 2018. **DOI: 10.1109/TED.2018.2834979.**

III. List of Publications after joining IIT Hyderabad (Without any student as co-author):

[19] Mohit Bajaj, **Kaushik Nayak**, Suresh Gundapaneni, and V. Ramgopal Rao, "Effect of Metal Gate Granularity Induced Random Fluctuations on Si Gate-All-Around Nanowire MOSFET 6-T SRAM Cell Stability," ***IEEE Transactions on Nanotechnology***, vol. 15, no. 2, pp. 243 - 247, March 2016. **DOI: 10.1109/TNANO.2016.2515638.**

IV. List of Publications before joining IIT Hyderabad (During Ph.D. Research):

[20] **Kaushik Nayak**, Samarth Agarwal, Mohit Bajaj, K. V. R. M. Murali, and V. Ramgopal Rao, "Random Dopant Fluctuation Induced Variability in Undoped Channel Si Gate All Around Nanowire n-MOSFET," **in** ***IEEE Transactions on Electron Devices***, vol. 62, no.2, pp. 685 – 688, February 2015. **DOI: 10.1109/TED.2014.2383352.**

[21] Aniruddha Konar, John Mathew, **Kaushik Nayak**, Mohit Bajaj, Rajan Pandey, Sajal Dhara, K.V.R.M. Murali, Mandar Deshmukh," Carrier Transport in High Mobility InAs Nanowire Junctionless Transistors," ***ACS Nano Letters***, February 2015.
DOI: https://pubs.acs.org/doi/10.1021/nl5043165.

[22] **Kaushik Nayak**, Samarth Agarwal, Mohit Bajaj, Philip J. Oldiges, K. V. R. M. Murali, and V. Ramgopal Rao, "Metal gate granularity induced threshold voltage variability and mismatch in Si gate-all-around nanowire n-MOSFETs,"

in IEEE Transactions on Electron Devices, vol. 61, no. 11, pp. 3892 – 3895, November 2014. **DOI:** **10.1109/TED.2014.2351401.**

[23] **Kaushik Nayak**, Mohit Bajaj, Aniruddha Konar, Philip J. Oldiges, Kenji Natori, Hiroshi Iwai, K. V. R. M. Murali, and V. Ramgopal Rao, "CMOS logic device and circuit performance of Si gate all around nanowire MOSFET," *in* ***IEEE Transactions on Electron Devices,*** *vol. 61, no. 9, pp. 3066 – 3074, September 2014.* **DOI:** **10.1109/TED.2014.2335192.**

[24] **Kaushik Nayak**, Mohit Bajaj, Aniruddha Konar, Philip J. Oldiges, Hiroshi Iwai, K. V. R. M. Murali, and V. Ramgopal Rao, "Negative differential conductivity and carrier heating in gate-all-around Si nanowire FETs and its impact on CMOS logic circuits," in ***JSAP Jpn. J. Appl. Phys.,*** *vol. 53, no. 4S, pp. 04EC16-1-04EC16-7, 2014.*
DOI: https://iopscience.iop.org/article/10.7567/JJAP.53.04EC16

[25] **Kaushik Nayak**, Mohit Bajaj, Aniruddha Konar, Philip J. Oldiges, Hiroshi Iwai, K. V. R. M. Murali, and V. Ramgopal Rao, "Negative differential conductivity in gate all-around Si nanowire FETs and its impact on circuit performance", ***Extended Abstracts of International Conference on Solid State Devices and Materials (SSDM),*** pp. 90-91, Fukuoka, Japan, Sep. 24-27, 2013. **DOI:** **10.7567/JJAP.53.04EC16.**

[26] Bharat Bhushan, **Kaushik Nayak** and V. Ramgopal Rao, "DC compact model for SOI tunnel Field-Effect Transistors ", in ***IEEE Transactions on Electron Devices,*** vol. 59, no. 10, pp. 2635 - 2642, October 2012. **DOI:** **10.1109/TED.2012.2209180.**

[27] Mrunal A. Khaderbad, **Kaushik Nayak**, M. Yedukondalu, M. Ravikanth, S. Mukherji, and V. Ramgopal Rao, *"Metallated Porphyrin Self Assembled Monolayers as Cu Diffusion Barriers for the Nano-scale CMOS Technologies"*, ***in 8th IEEE International Conference on Nanotechnology (IEEE NANO 2008),*** *pp. 167-170, Arlington, Texas USA, August 18-21, 2008.* **DOI:** **10.1109/NANO.2008.56.**

[28] **Kaushik Nayak**, Prasanna Kulkarni, Deepu A., V. Sitaraman, S. Punidha, A. Saha, M. Ravikanth, Sushanta Mitra, S. Mukherji, and V. Ramgopal Rao, *"Patterned Microfluidic Channels using Self-Assembled Hydroxy-phenyl Porphyrin Monolayer"*, ***in 7th IEEE International Conference on Nanotechnology (IEEE NANO 2007),*** *pp. 1235-1239, Hong Kong, August 2-5, 2007.* **DOI:** **10.1109/NANO.2007.4601406.**
